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A Versatile Carbon Fiber Cloth-Supported Au Nanodendrite Sensor for Simultaneous Determination of Cu(II), Pb(II) and Hg(II)

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Abstract: A carbon fiber cloth (CFC)-supported Au nanodendrite (AuND@CFC) sensor was used to simultaneously measure the concentrations of Pb(II), Cu(II) and Hg(II) in real water samples. The sensor had a large electrochemically active surface area due to the hierarchical nanodendrite structure and the formation of the nanodendrite structure on the CFC was accomplished in just 5 minutes by single-step electrodeposition. After the optimization of important experimental conditions, such as pH and pre-concentration time, the achieved limit of detections (LODs) in the measurements of Pb(II), Cu(II) and Hg(II) were 0.15, 0.07, and 0.13 ppb, respectively, which were superior and comparable to those reported in previous studies. The electrochemical responses of each analyte (concentration: 2.0 ppb) were not influenced when the concentrations of interferants (Cr(IV), Ni(II), Mn(II), Zn(II), Al(III), Ca(II) and Mg(II)) were even 500-fold greater (1.0 ppm); while only the signal of Cu(II) decreased when the Ni(II) concentration were 4.0 ppm (2000-fold larger). When eight separately prepared AuND@CFC sensors were used to measure the samples, the relative standard deviations (RSDs) of the peak intensities of these metals were lower than 5.0%, thereby indicating the superior sensor-to-sensor reproducibility. When real river and lake water samples were analyzed, the determined concentrations were similar to those measured with ICP-MS and the recoveries ranged from 93.2%–118.7%. Overall, the proposed sensor is versatile enough to be incorporated into a portable analytical system for on-site detection of Pb(II), Cu(II) and Hg(II) in real field samples.

Keywords: Carbon fiber cloth • gold nanodendrite • simultaneous determination • heavy metals • trace levels

As generally known, the family of heavy metal ions, such as Pb(II), Cu(II) and Hg(II) is highly toxic and causes diverse health problems like brain function deprivation and Wilson's diseases [1,2]. For this reason, the determination of these metal ions concentrations in water is necessary for the assessment of environmental conditions as well as for the protection of human health. For this purpose, atomic spectroscopy-based methods, such as inductively coupled plasma mass spectrometry (ICP-MS) and inductively coupled plasma atomic emission spectroscopy (ICP-AES), have been preferentially used while various types of electrochemical sensors that are able to simultaneously measure them have also been developed for simple and fast analysis without extensive sample pretreatment. The sensors developed for this purpose and the corresponding analytical performances are summarized in Table 1.

As shown in Table 1, the reported limit of detections (LODs) were generally greater than 1.0 ppb while the sensors utilizing the graphene/CeO_2 hybrid nanocomposite nanoparticle [3], 2-benzothiazolethiol functionalized SBA-15 mesoporous silica (BTSBA)/graphite/polyurethane composite [4], nanofibrillated cellulose (NFC) [5], and the nickel tungstate-graphene nanocomposite particle [6], provided superior sensitivity, with LODs lower than 0.05 ppb. However, these materials required long synthesis times and/or complex steps in their preparation. More specifically, the hydrothermal synthesis of graphene/CeO_2 hybrid nanocomposite needed 8 hours at 80 °C and the preparation of the nickel tungstate-graphene nanocomposite sensor required approximately 24 hours including hydrothermal synthesis, overnight drying, and additional thermal treatment. In addition, at least 24 hours were necessary for the drying of the epoxy resin used to seal the BTSBA/graphite/polyurethane composite and overnight drying was required for the film deposition of NFC gel on a glassy carbon electrode. Therefore, an

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Table 1. List of publications reporting electrochemical sensors that simultaneously measure Pb(II), Cu(II) and Hg(II) concentrations in samples. The employed sensor materials and the corresponding limit of detections (LODs) are also summarized.

Electrode	LOD (ppb)			Ref
	Cu(II)	Pb(II)	Hg(II)	
Calixarene bulk modified screen-printed electrode	40	38	48	[7]
Modified carbon paste electrode	0.51	1.04	1.60	[8]
EDTA_PANI/SWCNTs nanocomposite modified electrode	5.1	342	136	[9]
Mercaptoethanesulfonate (MES) monolayer modified Au nanoparticle electrode	0.145	0.156	0.131	[10]
Reduced graphene-oxide with L-cysteine modified glassy carbon electrode	0.261	0.416	1.113	[11]
Graphene/CeO_2 hybrid nanocomposite modified glassy carbon electrode	0.010	0.022	0.055	[3]
SBA-15 nanostructured silica/graphite/polyurethane composite electrode	0.06	0.02	0.12	[4]
Bismuth film modified graphite-polyurethane composite electrode	0.17	0.23	0.5	[12]
Montmorillonite-calcium modified carbon paste electrode	0.75	0.3	1.05	[13]
Nanofibrillated cellulose modified electrode	0.03	0.1	1	[5]
N-doped graphene modified electrode	0.32	1.04	10	[14]
Nickel tungstate–graphene nanocomposite electrode	0.013	0.04	0.06	[6]
Disposable screen-printed electrode	0.5	0.3	0.9	[15]
Screen-printed gold electrode	1	1.2	1.7	[16]
Ti_3C_2 MXene modified glassy carbon electrode	2.05	8.5	26	[17]
Vibrating gold microwire electrode	0.4	0.2	0.07	[18]
Zinc oxide-reduced graphene oxide nanocomposite electrode	2.56	6.21	12	[19]
Au nanoparticle modified glassy carbon electrode	19.2	62	60	[20]
2-aminothiazole functionalized microspheres modified carbon paste electrode	2.8	4.5	12.3	[21]
Organofunctionalized SBA-15 nanostructure silica modified carbon paste electrode	12.8	8.3	80.2	[22]
2-benzothiazole-2-thiol organofunctionalized SBA-15 silica printed electrode	1.92	13.5	9.2	[23]
Graphene oxide doped diamino-terthiophene-modified electrode	0.4	1.9	0.7	[24]
Boron doped diamond electrode	0.1	2	0.7	[25]
Chitosan-modified screen-printed carbon electrode	5	3.4	2	[26]
Graphene oxide on 4-aminophenyl modified gold electrode	1.7	0.3	1.7	[27]
Palladium nanoparticle/porous activated carbon (Pd/PAC) on glassy carbon electrode	4.22	10.35	10.8	[28]
Reduced graphene oxide/spongy Au nanoparticle electrode	1.75	2.63	4.2	[29]
Antimony-graphene oxide modified screen-printed carbon electrode	3.84	5.39	13.2	[30]
Neutral ionophores/nafion coated glassy carbon electrode	0.64	2.07	2	[31]

electrochemical sensor that is sensitive, easy-to-prepare, and reproducible is greatly beneficial for practical application, such as on-site sample analysis.

In this context, the carbon fiber cloth (CFC)-supported Au nanodendrite (AuND@CFC) demonstrated for a surface enhanced Raman scattering (SERS) substrate as well as an electrochemical sensor by our group [32] would be a strong potential candidate since it has a large electrochemically active surface area due to the hierarchical nanodendrite structure and the formation of nanodendrite on the CFC was accomplished in just 5 minutes by simple one-step electrodeposition. In addition, the porous structure of CFC provided a large internal volume to host many nanodendrites, and it also helped to rapidly bring analytes inside and into contact with the Au surface for the electrochemical reaction. Therefore, the AuND@CFC sensor has been employed to simultaneously detect Pb(II), Cu(II) and Hg(II) in aqueous samples, and important factors related to electrochemical responses, such as pH, pre-concentration time, and interferences, were examined. Furthermore, the sensor-to-sensor reproducibility was evaluated by observing peak intensities acquired using eight separately prepared sensors. Finally, the sensor was used to analyze real water samples collected from West Lake and Red River in Hanoi, Vietnam, and the subsequent results were compared with those obtained using ICP-MS.

The reagents used in this study were analytical grade and used without further purification. Double-distilled water was used for the preparation of all of the samples. Standard Pb(II), Cu(II) and Hg(II) solutions (1000 ppm, atomic absorption standard) were purchased from Merck. All of the reagents necessary for the construction of Au nanodendrites, $HAuCl_4$ (99.999 %), KI, H_2SO_4 and NH_4Cl were obtained from Sigma-Aldrich. CFC (20 cm × 20 cm × 0.36 cm) was purchased from Nara Celltech Corp., Korea. For the interference study, standard Al(III), Ca(II), Mg(II), Zn(II), Fe(III), Cr(III), Ni(II) and Mn(II) solutions (1000 ppm, atomic absorption standard, Merck) were diluted to the desired concentration levels.

The AuND@CFC sensor was prepared following the procedure reported in our previous publication [32]. In brief, a piece of CFC (6 × 6 mm, area: 0.72 cm^2) was cleaned using a mixture of acetone:water (1:1) in an ultrasonic bath and then activated by applying a potential of −1.0 V (vs Ag/AgCl) in a 0.5 M H_2SO_4 solution over 300 s. Next, the activated CFC was immersed in a solution containing 20 mM $HAuCl_4$, 1 mM KI, 5 M NH_4Cl and 0.5 M H_2SO_4 and a current of −70 mA was applied to it for construction of Au nanodendrites by electrodeposition. A scanning electron microscope (SEM) image of the

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built Au nanodendrite was obtained (S-4800, Hitachi, Japan) with an electron beam accelerated by a voltage of 15–20 kV to characterize the structure.

All of the electrochemical measurements were performed at room temperature (25 °C ±1) using a three-electrode system, in which an Ag/AgCl electrode and a Pt wire were used as the reference and counter electrode, respectively. A custom-made, multi-functional potentio-galvanostat manufactured by this research group (Vietnam Academy of Science and Technology, Hanoi, Vietnam) was used. It was equipped with 12-byte analog-digital and digital-analog converters with two operational amplifiers, and it provided the current resolution down to 0.008 nA.

In the simultaneous measurement of Pb(II), Cu(II) and Hg(II) in the samples, DPASV (differential pulse anodic stripping voltammetry) was used with a differential step of 0.005 V, a sampling time of 0.04 s, a pulse width of 0.08, and a pulse amplitude of 0.05 V. Pb(II), Cu(II) and Hg(II) were deposited by potentiostatic conditions at −0.35 V and voltammograms were recorded over a range of −0.35 to +0.7 V vs Ag/AgCl. A 0.1 M KCl/HCl solution was used as a supporting electrolyte in all of the measurements.

Real water samples were collected from two locations: Red River (a main northern river flowing through Hanoi) and West Lake (the biggest lake) in Hanoi, Vietnam. The sampled water was filtered through a Whatman GF/F glass fiber filter (45 µm) to remove any suspended particulates. Finally, the filtered solutions were preserved by adding HCl until the pHs of the samples were 1.0 and stored at 4 °C before analysis.

Before electrochemical measurements using AuND@CFC, its structure was examined by observing the corresponding SEM image (refer to Figure 1). As stated earlier, the Au nanodendrites shown in the figure were easily constructed via simple electrodeposition over 5 minutes, and their structures were apparently dendritic without aggregation or agglomeration. Therefore, a large surface area was surely available in the AuND@CFC sensor. The nanodendrites were also gently stacked in a three-dimensional domain with sample volume among them. This potentially made the analytes easily accessible to the surfaces of internally-located nanodendrites, thereby allowing maximal use of the available Au surface to secure the sensitivity. The optimal pH of a sample and pre-concentration time were initially searched. For this purpose, a sample containing 10 ppb Pb(II), Cu(II) and Hg(II) in an electrolyte of 0.1 M KCl/HCl was used. Using the AuND@CFC sensor shown in Figure 1, the sample was measured by changing the sample pH from 1.0 to 4.0.

Figure 2 (a) shows the voltammograms of the sample under the variation in pH. First of all, the oxidation peaks of Pb(II), Cu(II) and Hg(II) at −0.13 V, 0.32 V and 0.45 V, respectively were clearly observed and they did not overlap with each other. The peak intensities of these analytes were greatest at a pH of 1.0 and concurrently decreased with the elevation in pH up to 4.0 (refer to the inset). No noticeable peaks were observed when the pH was greater than 4.0. It can be explained that, as indicated in [33], acid dissociation constant ($-\log K_a$) for aqueous metal ions in solution follow the reaction:

$$M^n + H_2O \overset{K_a}{\rightleftharpoons} MOH^{(n-1)} + H^+$$

It causes, in the media more basic than pH 3.0, the peak current for the metal stripping peaks to diminish in magnitude and in the more acid solutions, metal stripping peaks become increasingly sharper. These changes are attributed to the metal hydroxide formation in the more

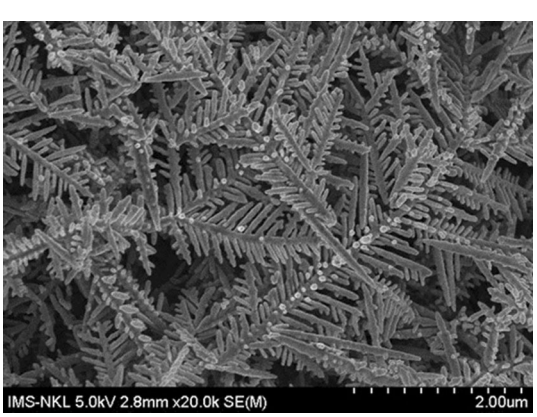

Fig. 1. SEM image of Au nanodendrite in AuND@CFC sensor.

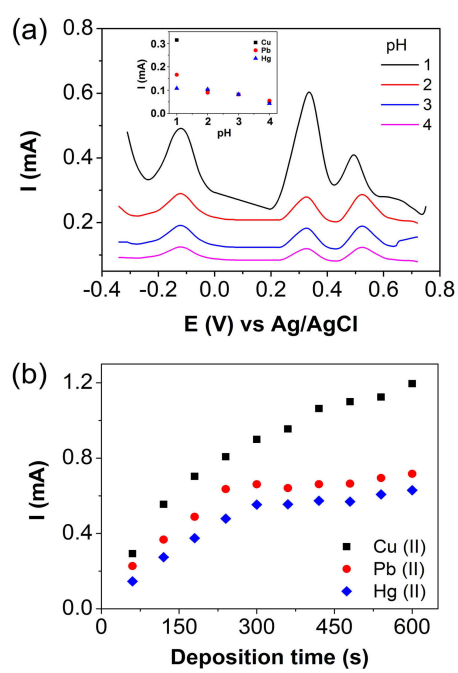

Fig. 2. (a) Stripping voltammograms of the samples containing 10 ppb Pb(II), Cu(II) and Hg(II) acquired with the variation in pH from 1.0 to 4.0. The inset shows the changes in peak intensities under these conditions. (b) Variations in the peak intensities of Pb(II), Cu(II) and Hg(II) when the pre-concentration time changes from 60 to 600 s.

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basic media. It is in agreement with previous publications [34,35]. Therefore, the pHs of subsequent samples were maintained at a pH of 1.0 for further electrochemical measurements.

Pre-concentration of analytes on an electrode surface is advantageous to enhance sensitivity. Therefore, using the same sample (10 ppb each), the pre-concentration time was changed from 60 to 600 s, and variations in individual peak intensities were observed. The peak intensities of Pb(II), Cu(II) and Hg(II) acquired with variation in the pre-concentration time are plotted in Figure 2 (b). In the cases of Pb(II) and Hg(II), the peak intensities gradually increased until the pre-concentration time was lengthened up to 300 s and no meaningful increase was observed afterward. Meanwhile, the intensity of the Cu(II) peak continuously increased over the tested range. Although a pre-concentration time longer than 300 s was advantageous to improve the sensitivity for Cu(II) measurement, the pre-concentration time was finally set to 300 s to maintain a balance between the sensitivities for all three analytes and the analysis time.

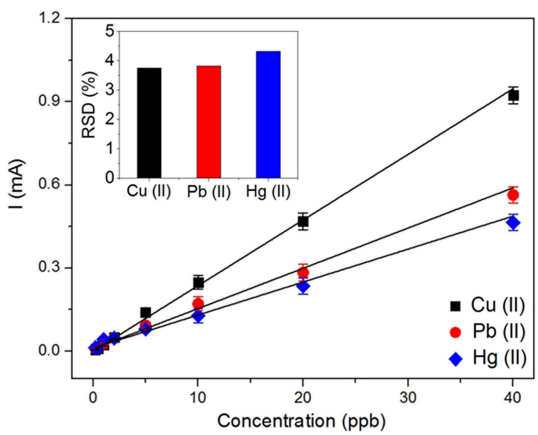

Fig. 3. Intensity variations in the Pb(II), Cu(II) and Hg(II) peaks when the samples in a concentration range of 0.2 to 40 ppb were measured. The error bars correspond to the standard deviations of peak intensities acquired with the eight separate sensors. The inset shows the average relative standard deviations (RSDs) of peak intensities in the measurements of each sample.

To evaluate the sensitivity, eight different samples containing the three metal ions in a concentration range of 0.2 to 40 ppb were prepared and measured using the AuND@CFC sensor at a pH of 1.0 with the pre-concentration time of 300 s. Eight separately prepared AuND@CFC sensors were used to measure each sample and also to examine the sensor-to-sensor reproducibility. Figure 3 shows the intensity variations of the Pb(II), Cu(II) and Hg(II) peaks according to the change in their concentrations. The error bars correspond to the standard deviations of peak intensities acquired with the eight separate sensors. In all three cases, the responses were clearly linear (R^2: 0.999) over the tested concentration range and the slope was relatively steeper for the measurement of Cu(II) as expected. The resulting LODs were very low and quite similar to each other: 0.15 ppb for Pb(II), 0.07 ppb for Cu(II) and 0.13 ppb for Hg(II). Although the LODs were slightly higher than those in the previous reports described in the introduction, the sensitive detection of these metal ions could be sufficiently realized using the AuND@CFC sensor. The inset shows the average relative standard deviations (RSDs) of the peak intensities in the measurements of each sample.

Table 2. Data of added and found concentrations and percent recoveries for each concentration used for the standard addition method.

Concentration of metal ions added (C_{add}, ppb)		Analyzed ($C_{anal.}$, ppb)			Average (ppb)			Stand Error			Stand deviation			Recovery ($C_{anal.}/C_{add}$) (%)		
		Cu(II)	Pb(II)	Hg(II)	Cu(II)	Pb(II)	Hg(II)	Cu	Pb	Hg	Cu	Pb	Hg	Cu	Pb	Hg
5	1st	4.76	4.79	4.68	4.89	4.93	4.79	0.094	0.106	0.085	0.163	0.183	0.147	97.8	98.8	95.8
	2nd	4.83	5.14	4.74												
	3rd	5.07	4.87	4.96												
10	1st	9.75	10.08	10.13	9.76	10.07	9.90	0.032	0.106	0.127	0.056	0.185	0.220	97.6	100.7	99.0
	2nd	9.82	9.88	9.69												
	3rd	9.71	10.25	9.89												
15	1st	14.72	14.53	15.32	14.55	15.09	15.19	0.111	0.287	0.163	0.192	0.497	0.282	97.0	100.06	101.3
	2nd	14.58	15.26	14.87												
	3rd	14.34	15.48	15.39												

Table 3. Cu(II), Pb(II) and Hg(II) concentrations in two real water samples determined by the AuND@CFC sensor and ICP-MS.

Unit: ppb

Sample	Cu(II)		Pb(II)		Hg(II)	
	AuND@CFC	ICP-MS	AuND@CFC	ICP-MS	AuND@CFC	ICP-MS
West Lake	23.9 ± 1.1	20.1 ± 0.1	11.3 ± 0.9	10.5 ± 0.1	< LOD	0.08 ± 0.01
Red River	9.2 ± 1.0	8.8 ± 0.1	5.9 ± 0.8	5.47 ± 0.1	< LOD	< LOD

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As shown, the RSDs for all three cases were lower than 5.0% even with the use of eight separate sensors, thereby indicating superior sensor-to-sensor reproducibility.

Next, the selectivity of measurements was evaluated by observing the electrochemical signals of 2 ppb Pb–Cu-Hg samples containing Cr(IV), Ni(II), Mn(II), Zn(II), Al(III), Ca(II) and Mg(II) as interferants. The concentrations of interferants ranged from 200 ppb to 4 ppm, equivalent to 100 to 2000 times greater than 2 ppb. The result of the interference test is shown in Figure 4. Percentage variations in the peak intensities in Figure 4 were calculated as below:

$$\frac{I^{Me}_{p\ (Pb,Cu,Hg)}*100}{I^{0}_{p\ (Pb,Cu,Hg)}}\ (\%)$$

Where:

$I^{Me}_{p\ (Pb,Cu,Hg)}$ is peak height of Pb(II), Cu(II) and Hg(II) in solution containing interferants.

$I^{0}_{p\ (Pb,Cu,Hg)}$ is peak height of Pb(II), Cu(II) and Hg(II) in solution without interferants.

The y-axis indicates the percentage of the variations in the peak intensities when the interferants were presented. In the cases of Pb(II) and Hg(II), there were no significant intensity changes even when the interferant concentrations increased to 4 ppm, 2000-fold greater than 2 ppb. It was found that the introduced error was only 3.8% in this case. Meanwhile, the intensity of the Cu(II) peak started to decrease by approximately 30% and 50% when the interferant concentrations were 1 and 4 ppm, respectively. When the concentration of Ni far exceeded the Cu concentration, such as 200:1, the formation of Cu–Ni film could occur under potentiostatic deposition. The formed Cu–Ni film inhibited the Cu stripping, thereby explaining the cause of the decreased signal [36–38].

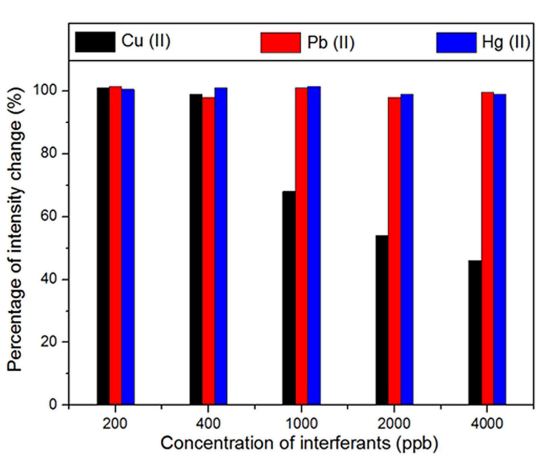

Fig. 4. Percentage variations in the peak intensities of 2 ppb Pb–Cu-Hg samples when the interferant (Cr(IV), Ni(II), Mn(II), Zn(II), Al(III), Ca(II), and Mg(II)) concentrations increased from 200 ppb to 4 ppm, equivalent to 100 to 2000 times greater than 2 ppb.

Finally, the AuND@CFC sensor was used to determine the concentrations of Pb(II), Cu(II) and Hg(II) in two real water samples (one collected from West Lake and the other from Red River, Hanoi, Vietnam). A standard addition method was used to determine the concentrations. Data of added and found concentrations and percent recoveries for each concentration used for the standard addition method is shown in Table 2. For comparison, the concentrations were also determined using ICP-MS. The results are summarized in Table 3. As shown, the concentrations of Hg(II) in the samples were very low, such as below the LOD, so the sensor was not able to detect Hg(II). The calculated Pb(II) and Cu(II) concentrations were similar to those measured with ICP-MS, and the recoveries ranged from 93.2% to 118.7%. These results indicate that the AuND@CFC sensor is sufficient for the potential analysis of real samples.

The analytical performances of the AuND@CFC sensor, such as the sensitivity and sensor-to-sensor reproducibility, were superior and also comparable to those reported in the previous studies described in Table 1. The preparation of the sensor was simple and easy-to-make by just one-step electrodeposition. This is a valuable merit when considering practical field application of the sensor. Therefore, a small and portable analytical device embedded with an AuND@CFC sensor as a disposable component is under development for on-site measurement of Pb(II), Cu(II) and Hg(II) concentrations in inland water samples.

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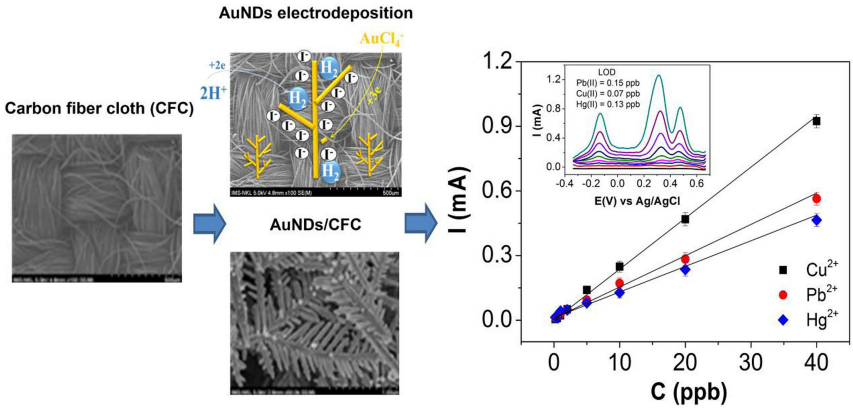
SHORT COMMUNICATION



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A Versatile Carbon Fiber Cloth-Supported Au Nanodendrite Sensor for Simultaneous Determination of Cu(II), Pb(II) and Hg(II)